

USDA-APHIS-Wildlife Services  
PERMANENT WOVEN WIRE FENCE SPECIFICATIONS  
for  
WOLF/LIVESTOCK CONFLICT ABATEMENT

47923 Hwy 112  
Ashland, WI 54806

**Technical Specifications** including:  
-Materials Specifications  
-Installation Specifications

**Materials Specifications**

**A. Wooden Products**

1. Posts must be straight.
2. Posts must be free from cracks (serviceable).
3. Posts must be dry (cured) (no sap).
4. Posts must meet diameter specification at smallest point. (IE: the 5" - 6" x 10' posts must be 5" - 6" in diameter on small end) and must be:
  - \* Gate, corner, and center in-line brace posts shall be 5 - 6 inches round x 10 feet long
  - \* Vertical and horizontal brace posts shall be 5 inches round x 10 feet long
  - \* Vertical and horizontal in-line brace posts shall be 5 inches round x 10 feet long
  - \* Line posts shall be 4 inches round x 10 feet long
  - \* Gate and corner assemblies shall be of "double H" construction as specified in the Installation Specifications
  - \* In-line brace assemblies shall be constructed as specified in the Installation Specifications
5. Posts pressure treated with pentachlorophenol, CCA, or ACZA must be treated to a minimum net retention of 0.4 pounds of preservative per cubic foot of wood and may be Douglas Fir, Western Larch, Western Hemlock, Red (Norway) Pine, Jack Pine, Ponderosa Pine, Southern Yellow Pine, Lodgepole Pine, or Northern White Cedar -or- Posts pressure treated with ACQ must be treated to a minimum net retention of 0.4 pounds of preservative per cubic foot of wood and may be either Douglas Fir or Southern Pine.
6. MINIMUM preservative penetration is ½ inch and 90% of sapwood for all species listed except: Southern yellow pine shall have a MINIMUM 2 ½ inch penetration and 85% of sapwood.

NOTE: Supplier must provide a signed statement that warrants the products meet the wooden product specifications 1-6 before acceptance. \*

**B. Galvanized Products:**

Galvanized metal products used as a part of the wolf fence must be class III galvanized steel and of sturdy construction to match the strength, durability, and function of the wolf fence.

**1. Woven Wire/Vertical Fence:**

Horizontal wires must be a minimum of 12.5 gauge, 179,000 minimum PSI tensile strength, minimum breaking load of 1,460 pounds, ASTM class III galvanized steel wire. Vertical stay wires must be a minimum of 12.5 gauge, 125,000 minimum PSI tensile strength, minimum breaking load 900 pounds, ASTM class III galvanized wire. All wire crossings must be knotted with a no-slip knot. Knot wire must be a minimum of 13 gauge, 69,000 minimum PSI tensile strength, minimum breaking load 400 pounds, ASTM class III galvanized wire. Horizontal wires shall have tension curves between every vertical wire. Mesh shall have a maximum of 6 inches space between all vertical wires. The horizontal line wires shall be graduated with a maximum wire spacing of 3 inches for the bottom 18 inches of the fence, and a maximum wire spacing of 7 inches for the top 21 inches of the fence, for a total mesh height of 75 inches.

2. Woven Wire/Horizontal Apron:

Horizontal wires must be a minimum of 12.5 gauge, 179,000 minimum PSI tensile strength, minimum breaking load of 1,460 pounds, ASTM class III galvanized steel wire. Vertical stay wires must be a minimum of 12.5 gauge, 125,000 minimum PSI tensile strength, minimum breaking load 900 pounds, ASTM class III galvanized wire. No-slip knot or slipknot may be used depending on cost and/or availability. Knot wire must be a minimum of 13 gauge, 69,000 minimum PSI tensile strength, minimum breaking load 400 pounds, ASTM class III galvanized wire. Mesh shall have a maximum of 6 inches space between all vertical wires. The horizontal line wires shall be graduated with a maximum wire spacing of 5 inches for the bottom 10 inches of the fence, and a maximum wire spacing of 7 inches for the top 14 inches of the fence, for a total mesh height of 42 inches. Gate aprons shall consist of a 42-inch-high livestock panel and will be hinged near the bottom of the gate. Aprons shall be fastened using galvanized hog rings of adequate size.

3. Wires:

- a.) Use 12.5 gauge, 200,000 minimum PSI tensile strength, ASTM class III galvanized wire for all brace wires.
- b.) Use polyethylene and stainless steel filament wire or other inexpensive poly-combination line for guide wires.

4. Brace pins:

All brace pins (5" & 10") shall be class III galvanized steel, stainless steel, or fiberglass.

5. Staples:

Staples shall be barbed, class III galvanized steel 1 3/4" in length.

6. Crimping sleeves:

Crimping sleeves can be either short or long in length, or equivalent (e.g., Gripple).

7. In-line strainers:

Galvanized in-line strainers shall be used on corner, gate, and in-line brace assemblies.

8. Gates:

Standard woven wire wolf fence gates shall be 6' tall mesh, less bottom clearance for opening. Frame bracing shall be a minimum 1 5/8" O.D. tube, .065 tube, 1.11 pounds per foot. Gate shall have cross bracing interior of the gate frame. Gates shall be fitted with a 1 5/8" swivel pin. Gates shall conform to galvanized products standard. Mesh shall meet or exceed woven wire standards. Standard gate width shall be 12', 16', 20', or 30' (20' gate shall consist of two 10' gates, 30' gate shall consist of two 15' gates). Standard number of gates is five gates of any size per wolf fence (NOTE: Additional gates may be installed at the sole expense of the landowner). An appropriately sized carabiner shall be included with the gate to fasten up the gate apron when opening and closing the gate.

NOTE: Supplier must provide a signed statement that warrants the products meet the galvanized products specifications 1-7 before acceptance. \*

9. Interior Electric Fence:

A single strand of 12.5 gauge, 200,000 PSI tensile strength, ASTM class III galvanized high tensile wire shall be installed at a height of 3' interior of the vertical mesh fence to prevent cattle from damaging the mesh fence. The wire shall be insulated with 6" offset screw-in ring insulators at line posts, and with porcelain bull nose insulators, donut insulators, or tube style insulators at corner posts, end posts, and gate posts. Wire shall be tightened using inline strainers and compression springs. Location and size of energizer will be determined during onsite pre-bid evaluations. 20,000 volt insulated wire shall be used to connect from the fence terminals on the energizer to the electric wire and grounding rods. Ground rods shall be 6' long, 1/2" diameter copper or galvanized metal rods. Standard number of ground rods is three. Wire shall be connected to the grounding rods using brass grounding clamps. 20,000 volt insulated wire housed in conduit shall also be used at gate openings. Galvanized hanger strap shall be used to fasten the conduit to the gate posts.

C. Aluminum Products:

1. Interior Electric Fence "Jumper Wire"

A single strand of 12.5 gauge, 430 lb breaking load aluminum fence wire shall be installed at a height of 3' interior of the vertical mesh fence to connect the electric fence of the main fence to the electric fence of the break-away fence.

D. Ditch crossings and topographic backfill materials:

Additional fencing materials must be installed at each site where the wolf fence crosses a ditch. The following materials are approved for filling the ditches or topographic depressions:

1. Shallow depression shall be filled with earth, gravel, galvanized panel, or mesh meeting or exceeding the woven wire specifications.
2. Break-away spans shall be woven wire meeting or exceeding the woven wire specifications, held in place with galvanized high tensile wire and rebar to secure to ditch bottom.

E. Other materials:

If a hole is augured into the ground for 6" X 10' corner, gate, and/or in-line brace posts, cement (provided by the fence installer and included as part of the installation costs) will be used to stabilize the post. Cement (provided by the fence installer and included as part of the installation costs) shall also be used to stabilize the base of other posts that cannot be set at the specified depth.

The USDA-APHIS-Wildlife Services nor these Materials Specifications endorse any brand of fence materials or products, any fence products company, or any fence installer.

The USDA-APHIS-Wildlife Services makes no warranty of any fence materials or products, any fence company or fence installer.

Subject to the Material and Installation Specifications, agreement(s) to purchase materials and installation services for each wolf fence are based on a low-cost estimation and the demonstrated ability of the fence product vendor and/or the installer to supply the quality and quantity of the materials needed and to erect the fence in the time limits specified.

\*Materials delivered which do not meet these Specifications shall be rejected. Independent testing labs may be consulted if it appears that materials provided fail to meet these specifications. Materials that fail such tests shall be rejected. Rejected materials will not be approved for use in the wolf fence, nor for payment. All costs of delivery, return, and/or testing of rejected materials may be billed to the fence company supplying the materials.

### **Installation Specifications**

The following Installation Specifications are a binding part of the wolf fence Agreement. Failure by the fence contractor to follow these Installation Specifications may result in non-payment.

If the wolf fence requires special features not detailed in these Installation Specifications, written Specifications for those features will be provided and those Specifications will become a binding part of this document.

If construction of the wolf fence requires modifications of these standard Installation Specifications, they must be approved in writing by the USDA-APHIS-Wildlife Services. Approved modifications are a binding part of this document. No modifications to decrease these specifications may be made without prior written approval from the USDA-APHIS-Wildlife Services.

If requested modifications involve any increase in acres enclosed, height, post frequency, post size, wire numbers, gates or any other increase in quantity, size, quality, etc. which adds to the cost of the fence, such request may be accepted provided:

- The result is still an effective wolf fence.
- The Landowner agrees in writing to assume full responsibility for the increased costs.

All material purchases will be made by the selected contractor. The Landowner is not authorized to purchase any materials to be included in this wolf fence.

A. General Information:

1. Safety warnings are printed in ALL CAPITALS throughout these Installation Specifications.
2. Fence building tips and reminders will be printed in *italics* throughout these Installation Specifications.
3. If a fence post cannot be set at the specified depth, the base of the post shall be stabilized with cement. This cement shall be provided by the fence installer and shall be included as part of the installation costs.

B. Site Preparation:

Survey work is sometimes required before any work can begin. Determining property lines is the Landowner's responsibility. Any costs incurred for any survey work or property line determinations are at the full expense of the Landowner.

Before actual fence construction can begin, the fence line will be cleared of trees, brush, weeds, rocks, debris, and old fences. The Landowner will clear a swath of at least 16' wide; this will provide adequate open area for future fence maintenance, including mowing along either side of the fence. Site preparation costs and any legal survey work will be the responsibility of the landowner.

A bulldozer is the best way to prepare the site for wolf fence installation. The site should be disced and dragged, providing a smooth run for the fence. All low spots must be filled with earth, gravel, or plugged with mesh to meet or exceed fence standards.

C. Corner, End, and Gate Opening Posts:

After the site has been adequately prepared, actual construction may begin. Set all of the 5" - 6" corner posts and end posts 44" deep, leaving 76" above ground. Set one 5" - 6" post on each side of each gate opening. Using the poly-twine, string a guide wire 1" off the ground from the center of the first gate opening post and on around the corners. *Remember: the wire will run on the inside of the completed fence.*

*The only exception*

*is an "inside corner", where the wire must run on the outside of the corner assembly.* This guide wire will mark the run of the entire fence. USE CAUTION WHEN INSTALLING THE WIRE; ENTANGLEMENT CAN BE DANGEROUS ESPECIALLY WHEN USING A PTO AUGER.

D. Gate, Corner, and End Assembly Installation:

*Corner, end, and gate assemblies are designed in a "double H" to provide the extra strength required for a tension fence.*

1. Gate Assembly Installation

a) Vertical Post Installation:

The gate assemblies will be arranged so that there is a "double H" on each side of the gate opening. The 5" - 6" gate post has already been set and tamped into place. Dig two post holes, placing the first hole 10' from the gate post and the second hole 10 feet beyond the first hole. Place a 5 - 6" vertical brace post into each hole so that the post is 44" deep and 76" above ground. DO NOT back fill or tamp them in at this time. All vertical posts in the gate assembly are spaced 10' apart. *Note: the guide wire runs from the center of the gate post to the inside of the vertical brace posts.*

b) Horizontal Brace Post Installation:

The horizontal brace posts are placed 62" from ground level. Place each 5" - 6" x 10' horizontal brace post as required, securing in place using 5" or 10" brace pins. Use 5" pins to secure the horizontal brace post to the 5" - 6" gate posts and to the second vertical post out from each corner. Use the 10" brace pins to fasten the horizontal brace post to the middle post of each double H.

Drill holes into each vertical post where the pins are to be inserted, at 62" above ground level. Drill a hole approximately 2.5" in length into each 5" - 6" gate post where the first brace is to be fastened. Drill a hole through the entire diameter of the next vertical post. Then drill another 2.5" hole into the second vertical post out from the gate post assembly. Finally drill a 2.5" hole into the center of both ends of each horizontal brace post.

Insert a 5" pin into the hole in the gate post and into the hole in the second vertical post. Insert a 10" brace pin into the middle post and drive it through until approximately 2.5" of the pins protrudes from each side of the 5" - 6" post. Slip a horizontal brace post onto the gate post pin, raising the brace into place and slipping the middle post pin into the other end of that brace. Repeat this process to construct the "Double H" assembly. After completing the double H construction, straighten the vertical posts, backfill, and tamp into place. If a hole is augured into the ground for 6" X 10' corner or gate posts, cement (provided by the fence installer and included as part of the installation costs) will be used to stabilize the post.

*Reminder: The wire will run from the center of the gate post and on the inside of both of the vertical posts.*

c) Diagonal Brace Wire Installation:

*A brace wire is required for each half of the “double H” assembly. 12.5 gauge high tensile wire is used for diagonal brace wires. Always construct the brace wire so that the top loop is pointing in the direction of the pull of the wire tension. Approximately 60' of wire will be needed for each 10' assembly.*

Beginning at the post assembly closest to the gate/corner post, loosely wrap the wire around the bottom of the gate post and around the top of the adjacent 5" – 6" vertical post twice, so that the ends of the wire meet in the middle of the span. Make a loop in one end of the wire using 1 long crimping sleeve, 3 short crimping sleeves, or equivalent (e.g., Gripple) crimping together using a crimping tool. Place the looped end through the ratchet base. Insert the other end of the wire into the hole in the ratchet spool. Lay a staple parallel with the post between the wires and the post. Secure the wires and staple laying against the post by loosely stapling 2 additional staples 3/4 deep into the post (do not pound these 2 staples completely into the post). Carefully tighten the brace wire with the ratchet.

F. Corner Assembly Installation:

The corner assemblies will be arranged so that there is a “double H” on each side of the 5" – 6" corner post. Corner assemblies are constructed in much the same way as the gate assemblies. The only difference is the post arrangement. In corner assemblies there is just one corner post, with a “double H” assembly on each side of the corner post.

G. End Assembly Installation:

The end assembly will consist of a single “double H”. End assemblies are constructed in the the same way as a gate assembly.

H. Line Post Installation:

*After all gate and corner assemblies have been completed, line post installation can begin. Use the guide wire to determine straight line post placement. Avoid curving fence line, or the addition of slight angles.*

Begin at any gate post assembly. Pace off distance of no greater than 25' and lay a line post to mark spot. Continue until all line posts are marking their proper position. *(In some topography, 20' line spacing may be necessary. Line posts are typically 4" x 10' wooden posts but can be up to 5" – 6" x 10' depending upon terrain and soil type. Proper line posts and spacing will be determined by USDA-APHIS-Wildlife Services).*

Dig holes or drive line posts into place, setting posts 44" deep and leaving 76" of post sticking out of ground. *Reminder: Wire goes on inside of post.* CAUTION: AVOID ENTANGLEMENT IN GUIDE WIRE.

I. In-line Brace Assembly Installation:

*In-line braces will be installed when stretching more than two rolls (660') of wire in a run. They cannot be used for corner or gate assemblies.*

In-line assemblies are constructed so that there is a 5" – 6" X 10' post set 44" deep and 76" sticking out of the ground. Ten feet to both sides of this post, a 5" – 6" X 10' post is set 44" deep and 76" sticking out of the ground. The brace wire is constructed in much the same way as on the corner/gate assembly, by loosely wrapping wire around the bottom of the 5" – 6" X 10' and around the top of the adjacent 5" – 6" vertical posts in either direction. Construction of the in-line brace assembly can be completed by following the horizontal brace post and diagonal brace wire installation procedures. If a hole is augured into the ground for 5" – 6" X 10' in-line brace posts, cement (provided by the fence installer and included as part of the installation costs) will be used to stabilize the post.

J. Woven Wire Installation:

Begin at any gate opening or end post assembly. Remove a minimum of one vertical wire from the end of the woven wire. Wrap every horizontal wire around one of the end posts and make a wire tie back to itself. *(Avoid damaging galvanized coating).*

Unroll the woven wire onto ground inside of fence posts. Attach additional woven wire as needed, using 1 long crimping sleeve, 3 short crimping sleeves, or equivalent (e.g., Gripple) per horizontal wire. Attach the 8' woven wire stretcher bar, to the woven wire, several feet beyond the next corner post. Using one or two hand winches, begin to lightly tension fence. *If run is short, flat, and clear, the wire mesh can begin to bring itself up. If not, prop wire temporarily as needed.*

Follow the fence line along temporarily placed woven wire to check for any dips or rises in the terrain resulting in gaps between ground level and fence wire. In all low spots, pull down bottom wire of mesh and staple temporarily into place. Along rises, lift top wire of mesh into place and staple temporarily into place. DO NOT staple tightly to line posts. Complete tensioning the fence, gently pulling the strainer bar until approximately half of the curve in the tension curve is removed. Go back to each post in run, and staple wire to each line post, placing a staple on every other horizontal wire. Use double staples on every wire on all corner posts to maintain tension. After woven wire has been secured to posts, carefully release tension on the wire, cut to size, and tie off joints with every horizontal wire around the end post.

The 42" apron will be installed on the outside of the vertical fence lying flat on the ground. If the apron is concaved, the apex should be facing upwards to ensure that the outer most edge of the apron is flush to the ground. The apron will be hog-ringed to the bottom horizontal wire of the vertical fence every 5' and will be "notched" around each post. The landowner will top seed, using a native grass seed mixture, areas that are disturbed to prevent soil erosion.

#### K. Gate Installation:

Attach hinges to gate. Measure the distance from the hinge to the gate bottom and add 6". On the gate post, mark and drill a hole for placement of the hinge bolt. Measure the distance from one hinge bolt to the next and drill another hole into gate post for placement of the second hinge bolt. *(These two holes should be in alignment to assure even gate placement)*. Place hinge bolts through holes in gate post, place washer and nut on both the outside and inside of bolt on either side of post. Adjust inner nuts to level of gate, and secure outside nuts. *(The bolt should now have a nut and washer on each side of the 6" post)*. Check for even gate swing, readjust as necessary. *(Distance from gate bottom to ground level must be as near to 6" as terrain allows but must not exceed 6" at any point. A hinged 42" apron constructed from livestock panel must be installed so that when the gate is closed the apron will prevent predators from digging under the gate. This apron must be hinged so it can be raised off the ground to facilitate opening and closing of the gate. An appropriately sized carabiner should be attached to the gate so that the apron can be fastened up while opening and closing the gate)*.

#### L. Interior Electric Fence Installation:

Fasten bullnose insulators, donut insulators, and/or tube style insulators as appropriate to all end posts, gate posts, and corner posts at a height of 3' so that the insulators project 6" out from the woven wire. Fasten 6" offset screw-in ring insulators at a height of 3' to all line posts; tube style insulators may be used on "double H" assemblies on each side of corner posts at inside corners. Beginning at any end post or gate post, attach the high tensile wire to the appropriate style insulator using 2 crimping sleeves. Unroll the high tensile wire along the fence line, securing the wire to the insulators that are fastened to corner posts as they are encountered. Attach additional wire as needed using 3 short crimping sleeves or equivalent (e.g., Grippler). Secure the wire to the appropriate style insulator at the next end post or gate post encountered. Use standard fencing practices to ensure that the high tensile wire does not contact the end post assemblies, corner post assemblies, gate post assemblies, or woven wire, creating a ground fault. Attach the wire to the 6" offset screw-in ring insulators. Install one inline strainer in the middle of every fence span. An inline strainer is capable of tensioning 4,000' of wire; however, every friction point (corner, sweep, dip, rise, etc.) reduces the pulling capability of a strainer by 500'. Install additional inline strainers in fence spans exceeding 4,000' or where friction points have reduced the pulling capability of the strainer to less than the distance of the fence span. Use a compression spring in combination with an inline strainer in fence spans less than 440'. Fasten the high tensile wire to the springs and strainers using 2 crimping sleeves. Tension the inline strainers to 250 pounds. To traverse gate openings, place insulated electric wire in conduit and bury to a depth of at least 2'. Bend at least 6" of the conduit and insulated wire downward to prevent water from getting in the conduit. Fasten the conduit to the gate posts using hanger strap. Connect the insulated wire to the high tensile wire. Install the ground rods perpendicular and external to the woven wire fence with a distance of 8' between ground rods. Connect the fence energizer to the high tensile wire and ground rods using the insulated electric wire. Fasten the electric wire to the ground rods using the brass grounding clamps. *Reminder: Wire goes on inside of posts.*

M. Break-Away Fence Installation:

a) Woven Wire:

A break-away woven wire fence must be installed in the location(s) indicated on the schematic where the wolf fence crosses a drainage or ephemeral stream, relieving water pressure on and preventing damage to the main fence during floodwater events. This fence must adhere to the previously described installation specifications required of the main fence. The main fence must terminate in an end post assembly on each side of the drainage. The break-away fence must be installed exterior to the main fence and must begin and terminate in end post assemblies. Each break-away end post assembly must overlap each main fence assembly by 5' and there must be no more than a 4" gap between the break-away fence assembly and main fence assembly overlap.

b) Interior Electric Fence:

A break-away electric fence must be installed on the break-away woven wire fence. The electric fence must adhere to the previously described installation specifications. It must begin and terminate on the break-away woven wire end post assemblies. The electric fence on the main fence must terminate at the end post assemblies of the main woven wire fence on each side of the drainage. An aluminum "jumper wire" must connect the main electric fence to the break-away electric fence on each side of the drainage. The "jumper wires" must be tightly fastened to the break-away electric fence and lightly fastened to the main electric fence so that they can pull off the main electric fence during floodwater events. The "jumper wires" shall be installed at a height of 3' and shall have 1' of slack in them. Alternative methods of connecting the main electric fence to the break-away electric fence that are equivalent or superior to these specifications in function may be implemented upon approval by USDA-APHIS-Wildlife Services.

N. Ditch Crossing:

Additional fencing must be installed at each site where the wolf fence crosses a ditch. Several methods may be used to fill gaps in between the bottom of the fence and ground level. The most cost-effective method will be approved by the USDA-APHIS-Wildlife Services.

1. Shallow depressions shall be filled with earth, gravel or mesh meeting or exceeding the wire specification.
2. Deeper ditches may be
  - a. Seasonally flooded or
  - b. May require a break-away span.
    1. Cut woven-wire to a shape that will completely close the ditch.
    2. Attach lightly to a guide wire.
    3. Pin bottom of wire securely to ditch bottom with rebar.
    4. *Keep all vegetation away from woven wire barrier to provide for adequate water flow.*
  - c. May require the installation of a culvert and fill material to facilitate water flow and provide for a flat surface for the fence and apron. Rebar will be used to prevent predators from accessing the pasture through the culvert.
  - d. Both a, b, and & c.

O. Special Considerations:

1. If applicable, livestock contained in pastures directly adjacent to and on the exterior of the wolf fence must be excluded from the apron so as to prevent it from being damaged by trampling until such a time as the apron has become integrated with (i.e., "grown into") the landscape. The Landowner will construct an electric fence consisting of a minimum of one wire at least 4' away from the wolf fence; the wolf fence may not be used to support the electric fence.
2. The Landowner will adhere to all relevant site preparation requirements as described in **Installation Specifications, (B) Site Preparation**.

P. The USDA-APHIS-Wildlife Services nor these Technical Specifications endorse any brand of fence materials or products, any fence company, or any fence installer. The USDA-APHIS-Wildlife Services make no warranty of any fence materials, products, fence company or installer.

Companies providing installation of the fence may be required to warrant in writing that they will comply with the Installation Specifications. Installation failing inspection may be grounds for non-payment when installation has not complied with specifications.